

Replication Plan

Clustering Huge Protein Sequence Sets in Linear Time

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Auto-generated Replication Blueprint

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Paper Overview

This paper presents Linclust, a linear-time clustering algorithm for protein sequence databases, implemented within the MMseqs2 software suite. Unlike CD-HIT and UCLUST which scale quadratically or worse, Linclust achieves $O(N)$ runtime by using a k -mer-based pre-filtering step to identify candidate cluster members, followed by alignment-based verification. The paper benchmarks Linclust on UniProt (hundreds of millions of sequences), demonstrating orders-of-magnitude speedup over competitors while maintaining comparable clustering quality.

Detailed Workflow

Phase 1: Software Installation

- 1.1 Install MMseqs2 (includes Linclust) from GitHub or conda.
- 1.2 Install competitor tools for benchmarking: CD-HIT, UCLUST/USEARCH, DIAMOND, MASH.
- 1.3 Verify installations with small test datasets.

Phase 2: Dataset Acquisition

- 2.1 Download UniProt/UniRef datasets.
 - UniRef100, UniRef90, UniRef50 FASTA files.
 - Specific release version as used in the paper (if identifiable).
- 2.2 Prepare subsets for scaling experiments (e.g., 1M, 10M, 50M, 100M, full database).

Phase 3: Clustering Experiments

- 3.1 Run Linclust at multiple sequence identity thresholds (e.g., 30%, 50%, 70%, 90%).
 - `mmseqs linclust inputDB outputDB tmp -min-seq-id 0.3`
 - Record wall-clock time, peak memory usage, number of clusters.
- 3.2 Run competitor tools on the same datasets with equivalent parameters.
- 3.3 Scaling experiments.
 - Run on increasing dataset sizes; plot runtime vs. N .
 - Verify linear scaling of Linclust vs. super-linear scaling of competitors.

Phase 4: Quality Assessment

- 4.1 Cluster quality metrics.
 - Sensitivity: fraction of true homologs placed in the same cluster.
 - Specificity: fraction of cluster members that are true homologs.
 - Comparison against SCOP/SCOPe or Pfam domain assignments as ground truth.
- 4.2 Cluster size distribution analysis.
- 4.3 Alignment quality: sample and verify pairwise alignments within clusters.

Phase 5: Figure and Table Reproduction

5.1 Reproduce runtime vs. database-size scaling plots.

5.2 Reproduce quality comparison tables.

5.3 Reproduce cluster size distribution histograms.

Datasets

Dataset / Source	Type	Location / Notes
UniProt / UniRef	Public protein sequences	https://www.uniprot.org/downloads
SCOP / SCOPe	Structural classification	https://scop.berkeley.edu/
Pfam	Protein family database	https://www.ebi.ac.uk/interpro/ (merged into InterPro)
Metaclust	Metagenomic protein clusters	https://metaclust.mmseqs.com/

Models, Tools, and Software

Core Tools

Tool / Library	Version	Purpose / URL
MMseqs2 (includes Lin-clust)	latest	Primary clustering tool https://github.com/soedinglab/MMseqs2
CD-HIT	≥ 4.8	Competitor clustering tool https://github.com/weizhongli/cdhit
USEARCH / UCLUST	latest	Competitor clustering https://www.drive5.com/usearch/
DIAMOND	≥ 2.0	Fast protein alignment https://github.com/bbuchfink/diamond
MASH	latest	Fast genome/sequence distance https://github.com/marbl/Mash
Python / Matplotlib	≥ 3.8	Benchmarking scripts and visualization
GNU time / /usr/bin/time	system	Wall-clock time and memory measurement

Hardware Requirements

- For full UniProt: ≥ 128 GB RAM, ≥ 500 GB SSD storage.
- For subsets (10M sequences): ≥ 32 GB RAM.
- Multi-core CPU (MMseqs2 uses OpenMP for parallelism).